

```

libname fama 'H:\empiricalfinance\Fama&MacBeth\FAMA-Mecbeth';
/*****input
data*****/
PROC IMPORT OUT= FAMA.RAW
DATAFILE=
"E:\empiricalfinance\Fama&MacBeth\FAMA-Mecbeth\finalraw.csv" ;
RUN;

proc import out=fama.rf
datafile="E:\empiricalfinance\Fama&MacBeth\FAMA-Mecbeth\finalriskfree
.csv";
run;
/*****prepare data
*****/
/*delete void data*/
data fama.raw; /*If we change the content, we'd better change the dataset's
name*/
set fama.raw;

libname fama 'E:\empiricalfinance\Fama&MacBeth\FAMA-Mecbeth';
/*****input
data*****/
PROC IMPORT OUT= WORK.RAW
DATAFILE=
"E:\empiricalfinance\Fama&MacBeth\FAMA-Mecbeth\finalraw.csv" ;
RUN;
/*stkcd: stock code; Trdmnt: Trade date; Mretwd: stock's monthly return;
Cmretwdeq: market's monthly return*/
proc import out= fama.rf
datafile="E:\empiricalfinance\Fama&MacBeth\FAMA-Mecbeth\finalriskfree
.csv";
run;
/*Trdmnt: Trade date; Monrfret: Monthly risk-free return*/
/*****prepare data
*****/
/*delete void data*/
data fama.raw;
set work.raw;
if Mretwd^=. and Cmretwdeq^=.; run;
/*count length of each stock's time series as num, save stkcd, num in table
fama.number*/
proc sql noprint;
create table fama.number as
select Stkcd, count (stkcd) as num
from fama.raw

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group by stkcd;
quit;
data fama.raw2;
merge fama.raw fama.number;
by stkcd;
run;
/*only keep stocks with more than 19-year data, i.e. 12*19=228 months */
data fama.original;
set fama.raw2;
if num>=228;
run;
/*merge with risk-free rate*/

proc sql;
create table fama.neworiginal as
select original.*,rf.Monrfret
from fama.original full join fama.rf
on original.Trdmnt=rf.Trdmnt;
quit;
/*full join
http://blog.csdn.net/rongbo\_j/article/details/46352337*/
proc sort data=fama.neworiginal;
by stkcd trdmnt;
run;

/*trial for 2003 year*/

/*每个时间段(共计4个)分别运算portfolio formation period、estimation period、
testing period*/
data fama.period1a; /*提取portfolio formation period的数据*/
set fama.neworiginal;
if 1997<=year<=2000;
run;
proc sort data=fama.period1a;
by stkcd trdmnt; run;
proc reg data=fama.period1a noprint outest=fama.period1b; /*portfolio
formation period 时间序列回归求每只股票beta值*/
model Mretwd=Cmretwdeq;
by Stkcd;
run;
data fama.period1b;
set fama.period1b;
keep Stkcd Cmretwdeq;

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run;
proc univariate data=fama.period1b; /*求所有股票beta值的分位数*/
var Cmretwdeq;
output out=fama.a_pct pctlpre=p pctlpts=(5 10 15 20 25 30 35 40 45 50 55
60 65 70 75 80 85 90 95);
run;
data fama.period1c; /*根据beta值由小到大将股票分为20组*/
set fama.period1b;
if _n_=1 then set fama.a_pct;
if Cmretwdeq<=p5 then group1=1;
else if Cmretwdeq<=p10 then group1=2;
else if Cmretwdeq<=p15 then group1=3;
else if Cmretwdeq<=p20 then group1=4;
else if Cmretwdeq<=p25 then group1=5;
else if Cmretwdeq<=p30 then group1=6;
else if Cmretwdeq<=p35 then group1=7;
else if Cmretwdeq<=p40 then group1=8;
else if Cmretwdeq<=p45 then group1=9;
else if Cmretwdeq<=p50 then group1=10;
else if Cmretwdeq<=p55 then group1=11;
else if Cmretwdeq<=p60 then group1=12;
else if Cmretwdeq<=p65 then group1=13;
else if Cmretwdeq<=p70 then group1=14;
else if Cmretwdeq<=p75 then group1=15;
else if Cmretwdeq<=p80 then group1=16;
else if Cmretwdeq<=p85 then group1=17;
else if Cmretwdeq<=p90 then group1=18;
else if Cmretwdeq<=p95 then group1=19;
else group1=20;
run;
data fama.group1a; /*提取每一组的股票代码每组为一个投资组合*/
set fama.period1c;
keep group1 stkcd;
run;
proc sort data=fama.group1a;
by Stkcd group1;
run;
data fama.group1b; /*分组信息与收益信息相结合*/
merge fama.group1a(in=a) fama.neworiginal;
by Stkcd;
if a;
run;
data fama.group1e1; /*提取每一个投资组合estimation period的股票信息*/
set fama.group1b;

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if 2001<=year<=2003;
run;
proc sort data=fama.group1e1; /*求投资组合收益率Rp*/
by group1 year trdmnt;
run;
proc means data =fama.group1e1 noprint;
var Mretwd Cmretwdeg;
by group1 year trdmnt;
output out=fama.bb mean=Rp Rm;
run;
Ods Output Summary=fama.Summary; /*求Rp和Rm的标准差*/
Proc Means Data=fama.bb n mean std nmiss;
Var Rp Rm;
by group1;
Run;
Ods Output Close;
data fama.bb;
set fama.bb;
drop _TYPE_ _FREQ_;
run;
proc reg data=fama.bb noprint outest=fama.cc adjrsq; /*回归求投资组合的残
差平方根和调整R方*/
model Rp=Rm;
by group1;
run;

proc sort data=fama.group1e1;
by group1 stكد year trdmnt;
run;
proc reg data= fama.group1e1 noprint outest=fama.group1e2; /*estimation
period: 每一投资组合组内每股时间序列回归求个股beta值*/
model Mretwd=Cmretwdeg;
by group1 Stكد;
run;
data fama.group1e2;
set fama.group1e2;
cmretwdeg2=cmretwdeg*cmretwdeg;
run;

data r; set fama.group1b; where year=2004; run; /*对2001-2003年估计的beta,
对应合并到2004年的return*/
proc sort data=r; by group1 stكد; run;

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```

data r1; set fama.group1e2; keep group1 stkcd cmretwdeq cmretwdeq2 _RMSE_;
rename cmretwdeq=coeff cmretwdeq2=coeff2; run;
proc sort data=r2; by group1 stkcd; run;

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data r2; merge r r1(in=id); by group1 stkcd; if id; run;
proc sort data=r2; by group1 year trdmnt; run;
proc means data=r2 noprint; var coeff coeff2 _RMSE_ Mretwd cmretwdeq
Monrfret; by group1 year trdmnt;
output out=fama.group_2003 mean=betap betap2 avestd Rp Rm Rf;
run; /*这里对应的return是2004年的，对应的回归beta是2001-2003年估计的*/

```

```

proc sql; /*将 _RMSE_, _ADJRSQ_, Rp_StdDev, Rm_StdDev, betap, avestd等信息
汇总到一个table*/

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create table fama.oo as
select group1, _RMSE_, _ADJRSQ_
from fama.cc;
quit;
proc sql;
create table fama.pp as
select group1, Rp_StdDev, Rm_StdDev
from fama.summary;
quit;

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proc sql;
create table fama.qq as
select group1, betap, avestd
from fama.groupiv_2003;
quit;

```

```

data fama.ss1;
merge fama.oo fama.pp;
by group1;
run;
data fama.ss2;
merge fama.ss1 fama.qq;
by group1;
run;

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```

data fama.ss_2003 ;/*即原文中table2*/
set fama.ss2;
betap_StdDev=_RMSE_/ (sqrt(36) *Rm_StdDev);
ratio=_RMSE_/avestd;
run;

```

```

proc sort data=fama.group1b;by group1 stkcd trdmnt;run;
proc reg data= fama.group1b noprint outest=fama.group1t1;
/*testing period aa-aa+1年: 每一个投资组合组内每股时间序列回归求个股beta*/
where 2003<=year<=2004;
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;

data fama.group1b1; /*提取aa+1年收益率信息*/
set fama.group1b;
where year=2005;
run;

data fama.x1; set fama.group1t1; /*保留回归分析中的个股beta和均方根误差*/
keep group1 Stkcd Cmretwdeq _RMSE_;
rename Cmretwdeq=coeff;
run;

data fama.group1tmt1; /*与aa+1年月收益率信息整合以排除退市股票*/
merge fama.group1b1 fama.x1(in=id);
by group1 Stkcd;
if id;
run;

proc sort data=fama.group1tmt1;
by group1 year Trdmnt;run;

data fama.group1tmt1;
set fama.group1tmt1;
coeff2=coeff*coeff;
run;

proc means data=fama.group1tmt1 noprint; /*对每月投资组合中所有股票beta求平均
得到aa+1年每月每一投资组合的beta和均方根误差*/
var coeff coeff2 _RMSE_ Mretwd Cmretwdeq Monrfret;
by group1 year Trdmnt;
output out=fama.group_2004 mean=betap betap2 aveStd Rp Rm Rf;
run; /*此处return是2005年的, beta估计来自2003-2004年*/

proc reg data= fama.group1b noprint outest=fama.group1t2; /*testing
period aa-aa+2年: 每一个投资组合组内每股时间序列回归求个股beta*/
where 2003<=year<=2005;
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;

data fama.group1b2; /*提取aa+2年收益率信息*/
set fama.group1b;
where year=2006;
run;

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```

data fama.x2; set fama.group1t2; /*保留回归分析中的个股beta和均方根误差*/
keep group1 Stkcd Cmretwdeq _RMSE_;
rename Cmretwdeq=coeff;
run;
data fama.group1tm2; /*与aa+2年月收益率信息整合以排除退市股票*/
merge fama.group1b2 fama.x2(in=id);
by group1 Stkcd;
if id;
run;
proc sort data=fama.group1tm2;
by group1 year Trdmnt;run;
data fama.group1tm2;
set fama.group1tm2;
coeff2=coeff*coeff;
run;
proc means data=fama.group1tm2 noprint; /*对每月投资组合中所有股票beta求平均
得到aa+2年每月每一投资组合的beta和均方根误差*/
var coeff coeff2 _RMSE_ Mretwd Cmretwdeq Monrfret;
by group1 year Trdmnt;
output out=fama.group_2005 mean=betap betap2 aveStd Rp Rm Rf;
run; /*此处return是2006年的, beta估计来自2003-2005年*/

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proc reg data= fama.group1b noprint outest=fama.group1t3; /*testing
period aa-aa+3年: 每一个投资组合组内每股时间序列回归求个股beta*/
where 2003<=year<=2006;
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;
data fama.group1b3; /*提取aa+3年收益率信息*/
set fama.group1b;
where year=2007;
run;
data fama.x3; set fama.group1t3; /*保留回归分析中的个股beta和均方根误差*/
keep group1 Stkcd Cmretwdeq _RMSE_;
rename Cmretwdeq=coeff;
run;
data fama.group1tm3; /*与aa+3年月收益率信息整合以排除退市股票*/
merge fama.group1b3 fama.x3(in=id);
by group1 Stkcd;
if id;
run;
proc sort data=fama.group1tm3;

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```

by group1 year Trdmnt;run;
data fama.group1tm3;
set fama.group1tm3;
coeff2=coeff*coeff;
run;

proc means data=fama.group1tm3 noprint; /*对每月投资组合中所有股票beta求平均
得到aa+3年每月某一投资组合的beta和均方根误差*/
var coeff coeff2 _RMSE_ Mretwd Cmretwdeq Monrfret;
by group1 year Trdmnt;
output out=fama.group_2006 mean=betap betap2 aveStd Rp Rm Rf;
run; /*此处return是2007年的, beta估计来自2003-2006年*/

```

/*检验capm模型*/

```

data fama.pltest; /*将时间段1的estimation period和三个testing periods整合
*/
set fama.group_2003 fama.Group_2004 fama.Group_2005
fama.Group_2006; /*2001-2003年20个组合beta (对应的组合收益率是R2004),
2003-2004年20个组合beta (对应的组合收益率是R2005), 2003-2005年20组合beta (对
应的组合收益率是R2006), 2003-2006年20个组合beta (不对应, 后来没有用到) */
run;

```

```

data fama.pltest;
set fama.pltest;

drop _FREQ_ _TYPE_;
run;

data fama.pltestn;
set fama.pltest;
if year^=.;
run;

proc sort data=fama.pltestn; by Trdmnt; run;

```

```

proc reg data=fama.pltestn noprint outest=fama.plresulta tableout
ADJRSQ; /*panela*/
Model Rp=betap;
by Trdmnt;
quit;

proc reg data=fama.pltestn noprint outest=fama.plresultb tableout
ADJRSQ; /*panelb*/

```

```

Model Rp=betap betap2;
by Trdmnt;
quit;
proc reg data=fama.pltestn noprint outest=fama.plresultc tableout
ADJRSQ;/*panelc*/
Model Rp=betap aveStd;
by Trdmnt;
quit;
proc reg data=fama.pltestn noprint outest=fama.plresultd tableout
ADJRSQ;/*paneld*/
Model Rp=betap betap2 aveStd;
by Trdmnt;
quit;

data fama.result1a;/*提取回归系数*/
set fama.plresulta;
if _TYPE_='PARMS';
by trdmnt;
run;
proc sql;create table fama.yyy1a as
select distinct result1a.*,rm,rf
from fama.result1a left join fama.pltest on
result1a.trdmnt=pltest.trdmnt;
quit;
data fama.yyy1a;set fama.yyy1a;diff1=intercept-rf;diff2=rm-rf;run;/*计
算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/
data fama.stder1a;/*提取回归系数的标准误*/
set fama.Plresulta;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;
proc univariate data=fama.yyy1a;/*计算回归系数、R方、Rm、Rf和diff的均值、标
准差、方差和t值*/
var Intercept betap _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_1_a
MEAN=meanr0 meanr1 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 trr trm trf tdif1 tdif2;
quit;
data fama.stder1a;
set fama.stder1a;
intercept2=intercept*intercept;

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betap2=betap*betap;
run;
Ods Output Means.Summary=fama.table2_1_a; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stder1a mean ;
Var Intercept2 betap2;
Run;
Ods Output Close;

data fama.result1b; /*提取回归系数*/
set fama.plresultb;
if _TYPE_='PARMS';
by trdmnt;
run;
proc sql; create table fama.yyy1b as
select distinct result1b.*,rm,rf
from fama.result1b left join fama.pltest on
result1b.trdmnt=pltest.trdmnt;
quit;
data fama.yyy1b; set fama.yyy1b; diff1=intercept-rf; diff2=rm-rf; run; /*计算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/
data fama.stder1b; /*提取回归系数的标准误*/
set fama.Plresultb;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;
proc univariate data=fama.yyy1b; /*计算回归系数、R方、Rm、Rf和diff的均值、标准差、方差和t值*/
var Intercept betap betap2 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_1_b
MEAN=meanr0 meanr1 meanr2 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdr2 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varr2 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 tr2 trr trm trf tdif1 tdif2;
quit;
data fama.stder1b;
set fama.stder1b;
intercept2=intercept*intercept;
betap2=betap*betap;
betap22=betap2*betap2;
run;
Ods Output Means.Summary=fama.table2_1_b; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stder1b mean ;
Var Intercept2 betap2 betap22;

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Run;
Ods Output Close;

data fama.result1c; /*提取回归系数*/
set fama.plresultc;
if _TYPE_='PARMS';
by trdmnt;
run;

proc sql; create table fama.yyy1c as
select distinct result1c.*,rm,rf
from fama.result1c left join fama.pltest on
result1c.trdmnt=pltest.trdmnt;
quit;

data fama.yyy1c;set fama.yyy1c;diff1=intercept-rf;diff2=rm-rf;run; /*计算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/

data fama.stderr1c; /*提取回归系数的标准误*/
set fama.Plresultc;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;

proc univariate data=fama.yyy1c; /*计算回归系数、R方、Rm、Rf和diff的均值、标准差、方差和t值*/
var Intercept betap aveStd_1 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_1_c
MEAN=meanr0 meanr1 meanr3 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdr3 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varr3 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 tr3 trr trm trf tdif1 tdif2;
quit;

data fama.stderr1c;
set fama.stderr1c;
intercept2=intercept*intercept;
betap2=betap*betap;
aveStd_12=aveStd_1*aveStd_1;
run;

Ods Output Means.Summary=fama.table2_1_c; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stderr1c mean ;
Var Intercept2 betap2 aveStd_12;
Run;
Ods Output Close;

data fama.result1d; /*提取回归系数*/
set fama.plresultd;

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```

if _TYPE_='PARMS';
by trdmnt;
run;

proc sql;create table fama.yyyld as
select distinct resultld.*,rm,rf
from fama.resultld left join fama.pltest on
resultld.trdmnt=pltest.trdmnt;
quit;

data fama.yyyld;set fama.yyyld;diff1=intercept-rf;diff2=rm-rf;run; /*计
算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/

data fama.stderrld; /*提取回归系数的标准误*/
set fama.Plresultd;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;

proc univariate data=fama.yyyld; /*计算回归系数、R方、Rm、Rf和diff的均值、标
准差、方差和t值*/
var Intercept betap betap2 aveStd_1 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_1_d
MEAN=meanr0 meanr1 meanr2 meanr3 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdr2 stdr3 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varr2 varr3 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 tr2 tr3 trr trm trf tdif1 tdif2;
quit;

data fama.stderrld;
set fama.stderrld;
intercept2=intercept*intercept;
betap2=betap*betap;
betap22=betap2*betap2;
aveStd_12=aveStd_1*aveStd_1;
run;

ods output Means.Summary=fama.table2_1_d; /*求回归系数标准误的平方的均值*/
proc means data=fama.stderrld mean ;
var Intercept2 betap2 betap22 aveStd_12;run;
ods output close;

```